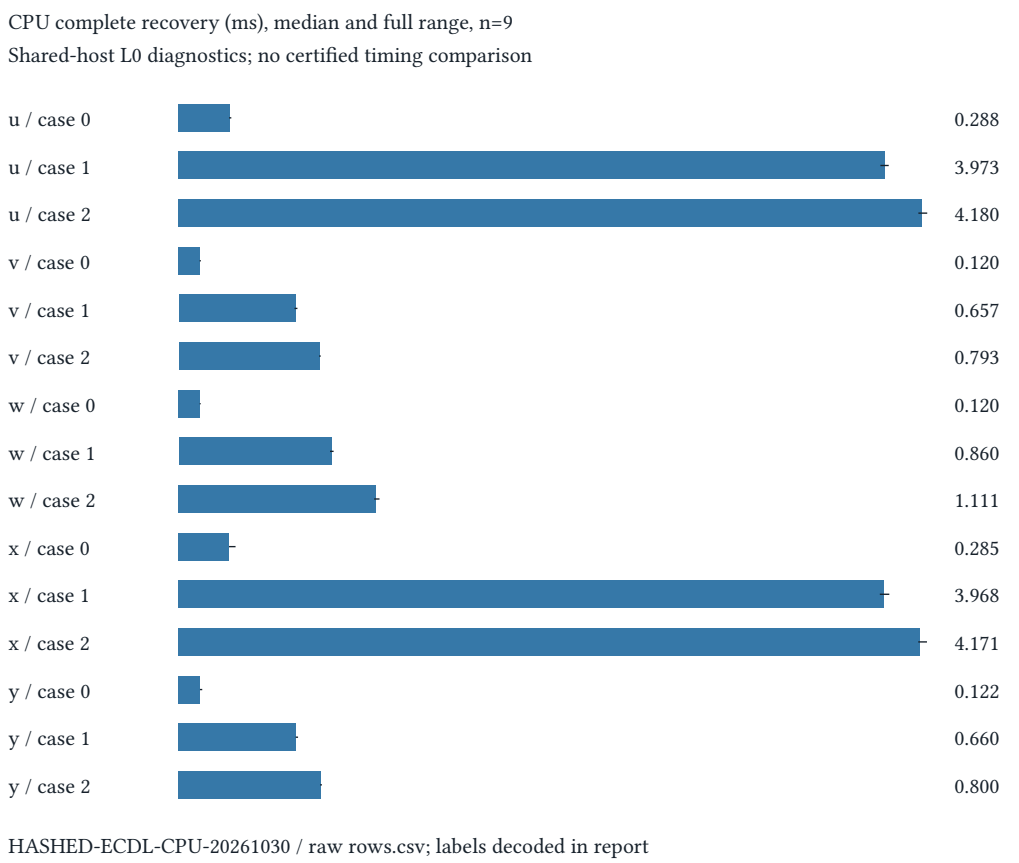
CPU direct projective witness comparison

Shared-host diagnostics only; no certified speedup or hypothesis promotion. Job `hashed-ecdl-cpu-round13-20261030`, October 30, 2026 UTC.

Original baseline u/x: `2bd4588fba60`. Candidate v: `92463171363c`; batch1 w. Previous candidate y: `308bf1a6a5bc`. One software thread, AMD EPYC9R45, 2 vCPUs / 4 GiB reserved, shared EKS VM (L0). All earlier controls remain present.

Fixture	Baseline u ms	Previous y ms	Candidate v ms	y/v
8-bit / 3 slots	0.287681	0.122430	0.119581	1.024
16-bit / 3 slots	3.972973	0.659942	0.657142	1.004
16-bit / 4 slots	4.180033	0.800253	0.792513	1.010

Medians from nine measured repetitions after one warmup. Baseline u/v ratios: 2.406, 6.046, 5.274. The further 10x target remains unmet.



Every oracle witness point and target is validated. The projective sum is compared by cross multiplication, avoiding its final inversion. Solver relation checks and both independent full certificate replays remain unchanged.

All 150 executions completed (135 measured,15 warmups). Full certificate bytes, scalar checks, replay and corrupted-scalar rejection pass. The complete interval charges setup, discarded trials, elimination, both checks and serialization. Process launch, archive I/O and negative controls are excluded.

All candidate/previous ranges overlap. Candidate v ranges (ms): 0.118280–0.125210, 0.652072–0.668372, 0.790512–0.799413; y: 0.120040–0.132980, 0.658352–0.671782, 0.795922–0.806052. Retain the removed inversion and exact checks; these timings do not certify improvement. All 31 release tests and strict Clippy pass. Ratios are derived from measured medians.

The export passes file-integrity/completeness checks with five-arm counts and its frozen analyzer identity. Earlier metadata/provenance corrections remain preserved in their own panels. Strict isolation, independent review, matched external/rho comparisons and larger-scale recovery remain unresolved. Protocol: `cf4d7243010c43da6325e869fd0db71f57b7016d67ebea44ee8906b64284e066`. Raw rows, certificates, build identity and checked export: `round13/results/`.